Solutions Questions

1. Define in your own words what is meant by the following terms:
    a. Solution:
    b. Solute:
    c. Solvent:
2. A teaspoon of coffee is dissolved in some boiling water to make a cup of coffee. Give the correct te each of the following:
    a. The coffee granules:
    b. The boiling water:
    c. The final cup of coffee:



3. Your 7-year-old sister is watching you put a teaspoon of sugar in a glass of water. She made a comment that the sugar has disappeared. How would you explain the 'disappearance' of the sugar to her?
4. Give an example of a solvent that is not water in a solution.
5. Draw a simple diagram to explain the difference between a 'concentrated' and a 'dilute' solution.
6. Describe how you would decant water from a mixture of sand and water.
7. Give an example of a mixture that you would use a kitchen sieve to separate. Describe how you would do this.
8. Draw a diagram that shows how you would use the process of filtration to separate sand from water. In your diagram label all the equipment you use and where the 'residue' and the 'filtrate' would be located.
9. Explain where and how a centrifuge could be used to separate a mixture.
10. Draw a diagram to explain how you would use the process of evaporation to remove salt from sea water.
11. Sketch a diagram to describe the process of distillation. In your sketch label the following: solution to be separated, condenser and distillate.
12. Give three examples of the use of distillation.
13. In what situation would chromatography be used to separate a mixture?
14. Filters are used in a number of parts of a vehicle engine, fo example air and oil filters. A feature of these filters is that they folded within their container. Explain why they are folded this way.



15. You are given a mixture of iron filings, copper sulphate and chalk. Draw a flow chart to describe how you will separate this mixture. You are free to use a range of materials that you would normally find in the lab. In preparation for your investigation, you tested the solubility of copper sulphate and chalk and found that the chalk was insoluble while the copper sulphate was soluble.
16. Explain which separation method would be used for the following situations. For each briefly state why each process works.
    a. Large rocks need to be separated from sand and very small rock in a limestone quarry.
    b. An industrial factory is belching out large amounts of black smoke into the atmosphere.

c. Plasma (fluid) needs to be separated from the cells in a blood sample.
d. A whisky manufacture wishes to separate alcohol from a mixture of water and other fermentation products.
e. You are trying to make sure that your video camera stays free of any moisture in the air while it is being stored in its bag.